
AWS CodeArtifact – Overview

What is AWS CodeArtifact?

AWS CodeArtifact is a **fully managed artifact repository** service that makes it easy to securely store, publish, and share software packages used in your development process.

Think of it like a private version of npm, Maven Central, PyPI, or NuGet — hosted inside AWS.

Key Features

Feature	Description
Supports Multiple Package Formats	npm (JavaScript), Maven (Java), PyPI (Python), NuGet (.NET), and more
Secure Package Storage	Uses AWS IAM and encryption for access control and data protection
Upstream Repositories	Connect to public repositories (e.g., npmjs.com, pypi.org)
Integration with DevOps Tools	Works with CodeBuild, CodePipeline, and third-party CI/CD tools
Version Control	Track and manage versions of your dependencies
Private Package Sharing	Share packages across teams or AWS accounts

Core Concepts

Concept	Description
Domain	A top-level container for multiple repositories. Think of it like a namespace.
Repository	Stores your actual packages. Can be linked to upstream repos.
Package	A versioned software component (e.g., an npm module or Maven artifact).

Upstream Repository

Connect to public repositories or other internal CodeArtifact repos to fetch packages.

Typical Use Cases

- Hosting **private npm/Maven/PyPI** packages
 - Creating a **centralized internal package registry**
 - Securing and caching packages from public repositories
 - Reducing build failures due to public registry downtime
 - Integrating with **CodeBuild** and **CodePipeline**
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Security & Access Control

- Access managed through **AWS IAM policies**
 - You can grant access to:
 - Individual users
 - CI/CD tools (e.g., CodeBuild)
 - Cross-account access (for multi-team organizations)
 - All data is encrypted with **AWS KMS**
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Pricing Overview

You pay for:

- **Storage:** Per GB stored per month
- **Requests:** Per read/write request

Free tier includes 2 GB of storage and 100,000 requests per month for the first 12 months.

[See full pricing →](#)



Sample Workflow: Using CodeArtifact with npm



Step 1: Create Domain & Repository

```
aws codeartifact create-domain --domain my-domain
```

```
aws codeartifact create-repository --domain my-domain --repository my-npm-repo
```



Step 2: Authenticate (npm login)

```
aws codeartifact get-authorization-token --domain my-domain --query authorizationToken  
--output text
```

Then configure npm:

```
npm config set
```

```
//my-domain-123456789012.d.codeartifact.us-east-1.amazonaws.com/npm/my-npm-repo/:_auth  
Token=<token>
```



Step 3: Install or Publish Packages

To **install**:

```
npm install my-package
```

To **publish**:

```
npm publish
```

CodeArtifact will store and version your packages automatically.

Integration with Other AWS Services

- **AWS CodeBuild:** Use CodeArtifact as a package source in builds.
 - **AWS CodePipeline:** Automatically pull packages as part of your CI/CD pipeline.
 - **CloudWatch Logs:** Monitor usage and errors.
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Summary

Benefit	Description
 Secure	Fully managed, IAM-based access
 Reliable	Cache or replicate public packages
 Integrated	Works with CI/CD pipelines and multiple languages
 Versatile	Host private packages or mirror public ones
